IPV4 → Limited.

Solution :→ (i) Private IP ✗

(ii) IPV6 ✓

(iii) Subnetting ✓

↓ (IPV4)

within a same network.

IPV6

32 bit.

→ netid

→ Hostid.

separate IP structure

CIDR Notation

(classless Inter-Domain Routing Protocol.)

VLSM

variable length subnet mask

classfull

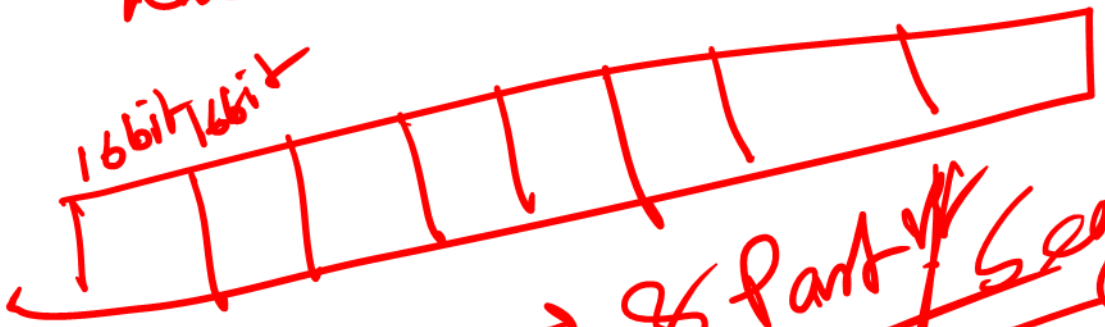
/ 23 25

IPV_5 ⇓ ✗

$IPV_4 \rightarrow IPV_6$ ✓

⇒ IPV_6 is not a successor of IPV_4.
⇒ IPV_6 has no backward Compatibility

128 bit.



16 bit ⇒ 8 Part / Segments.

1) Address space ⇒ 3.4×10^{38}

2) IPV_4 ⇒ Header Size ⇒ (20 - 60 Byte)
IPV_6 ⇒ " " ⇒ (40 Byte).

(iii) End to End Connectivity.

(iv) Auto Configuration. (DHCP6)

(v) Fast Forwarding.

(vi) No Broadcasting.

(vii) IP Security High.

(viii) Anycast support.

1 2 3 4 5 6 7 8

4bit 16bit 16bit 16bit

16 × 8 = 128bit

Hexadecimal Format.

0 - F.

0 - 9

A - F

(i) :

(ii) If two blocks are consequtively zero then we can use ::

A2B6 : 3257 : 0000 : 0000 : 123B : 342B : 5454 : 1212

A2B6 : 3257 : : 123B : 342B : 5454 : 1212

(iii) : 0000 : 0000 : 0000 : — : 0000 : 0000 : —

— : : — : 0000 : 0000 : —

(iv) 0025 : 0424 : 1010 : —

25 : 424 : 1010

40 Byte

A2B5
↳ Alphanumeric code.

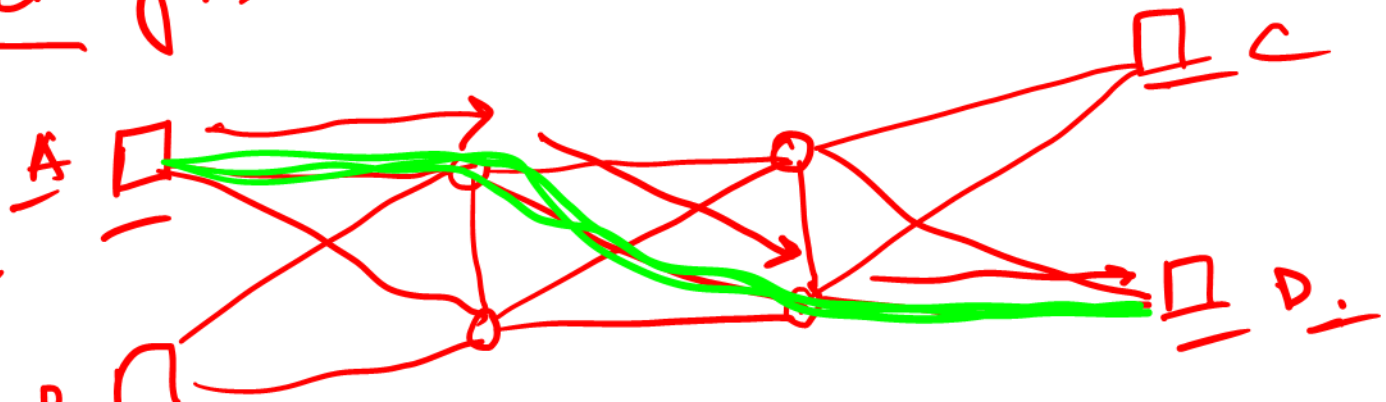
Switching Techniques.

- → Circuit switching
- → Message "
- → Packet "

Switching is a technique by which we can sent data among different route, i.e. it is a process of forwarding packets from one port to another.

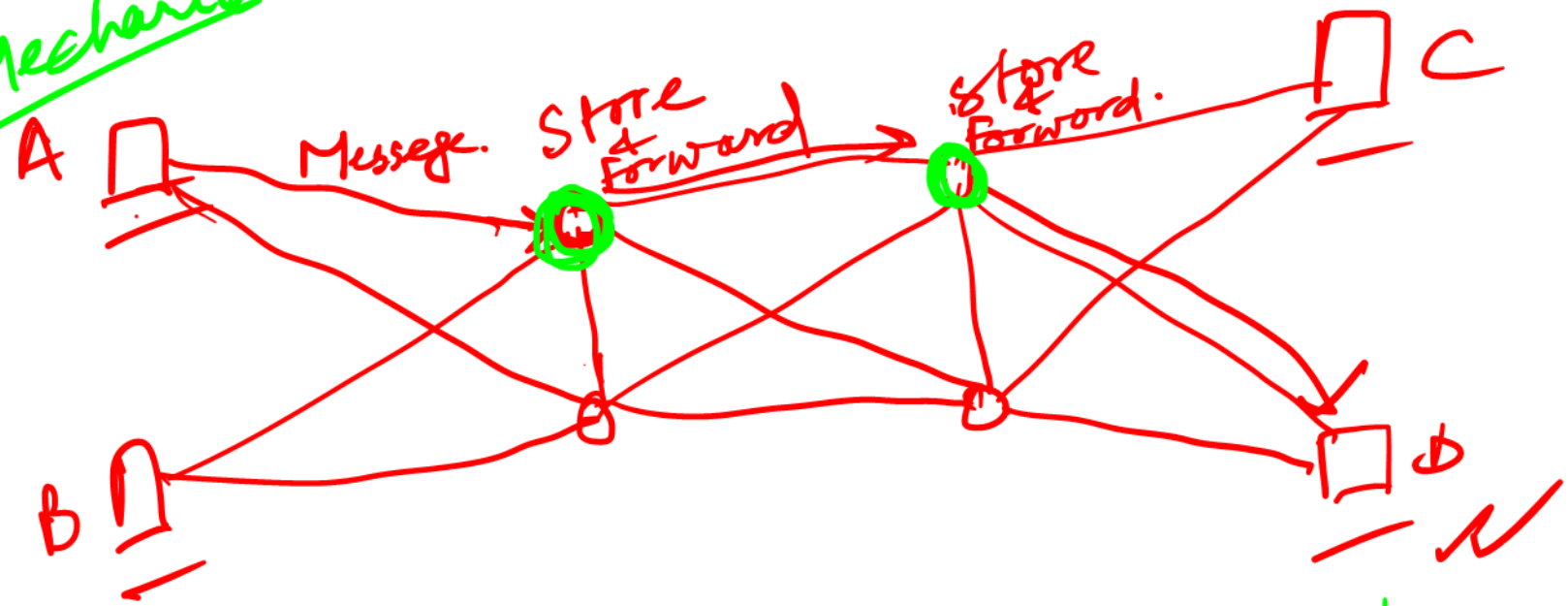
Circuit Switching :→

(i) circuit build (Dedicated) path
(ii) Data Transfer
(iii) Circuit Disconnection.



Message switching :→

Store & forward Mechanism.



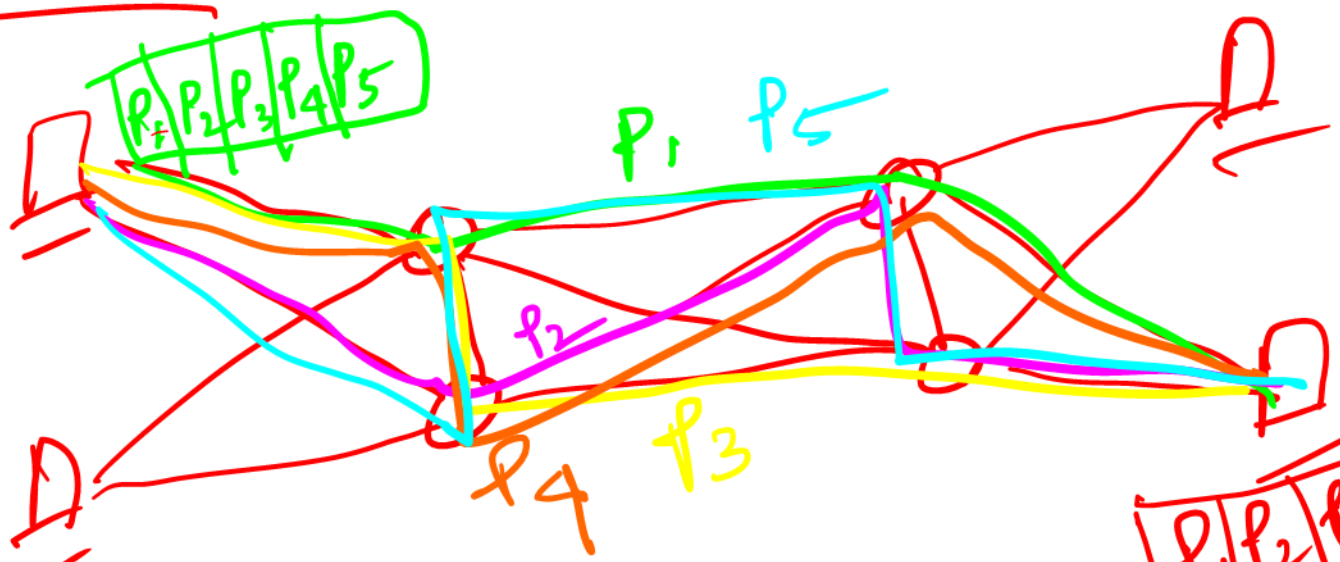
(i) Slow

(ii) Large capacity terminal.

Packet switching :→

Message

P_1	P_2	P_3	P_4	P_5



(i) Fast
(ii) Increases Bandwidth

(i) RAM
(ii) Processing Power.

P_1	P_2	P_3	P_4	P_5

Message

Reassemble.

HW :→ Comparative study of different switching Techniques.

Monday